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Chronic kidney disease stages 4–5: Patient management

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This is the third in a series of four articles which will discuss the cardiovascular implications within renal care. The aims of these articles are to highlight the important role that renal dysfunction plays in the care of cardiac patients. The first article addressed acute kidney injury and its impact on cardiac patients (Byrne and Murphy, 2008). The second article explored the role of chronic kidney disease (CKD) stages 1 to 3 and its close interrelationship with cardiovascular disease (CVD) (Murphy and Byrne, 2009). This third article in this series will explore CKD stages 4 to 5, while the final article will develop further the cardiac implications of renal replacement therapies.

This article will discuss the many physical symptoms which impact on individuals at CKD stages 4 to 5 including hypertension, dyslipidaemia, anaemia, potassium disorders, mineral and bone disorder, metabolic acidosis, gastrointestinal and dermatological disorders (*Figure 1*). It is vital that patients are prepared for renal replacement therapies at CKD stage 5 which include timely education including nutritional support along with psychological and social support.

The recent National Institute for Health and Clinical Excellence (NICE) guidelines identify that an average of

30% of people with advanced kidney disease are referred late to renal services from both primary and secondary care resulting in increased mortality and morbidity (National Collaborating Centre for Chronic Conditions (NCC-CC, 2008). Over 2% of the total NHS budget is spent on patients requiring renal replacement therapies. Patients at stage 4 and 5 CKD should therefore be referred for specialist assessment (with or without diabetes) (NCC-CC, 2008) (see *Table 1*).

The NICE guidelines (NCC-CC, 2008) further highlight that measures should be taken to identify progressive CKD. This includes obtaining a minimum of three estimated glomerular filtration rate (eGFR) measurements over a period of not less than 90 days. In individuals with a new result of a reduced eGFR, this eGFR should be repeated within a two-week timeframe to rule out causes of acute deterioration of GFR such as acute kidney injury or the instigation of a drug regime of angiotensin-converting enzyme (ACE) inhibitors or angiotensin receptor blockers (ARBs).

Furthermore, progression in CKD should be defined as a reduction in eGFR of more than 5 ml/minute/ 1.73 m² in

Table 1.

Individuals with CKD who should be referred for specialist assessment

- ♦ Higher levels of proteinuria unless known to be due to diabetes and already appropriately treated
- ♦ Proteinuria together with haematuria
- ♦ Rapidly declining estimate of GFR (eGFR) (more than 5 ml/minute/1.73 m² in 1 year, or more than 10 ml/minute/1.73m² within 5 years)
- ♦ Hypertension that continues to be poorly controlled in spite of using at least four antihypertensive drugs at therapeutic doses
- ♦ People with, or suspected of having, rare or genetic causes of CKD
- ♦ Suspected renal artery stenosis

NCC-CC, 2008

ABSTRACT

Chronic kidney disease (CKD) is classified into five stages with stages 1 to 3 identified as early CKD. This disease advances further in stage 4, while in stage 5 patients require renal replacement therapy (dialysis and transplantation) to survive. The risk factors for cardiovascular disease (CVD) are also the risk factors for CKD, while CKD is itself a key risk factor for CVD. A high proportion of patients with CKD will die from CVD. The nursing care of patients with stages 4–5 CKD includes the management of the many physical symptoms associated with these late stages such as hypertension, dyslipidaemia, anaemia, potassium disorders, mineral and bone disorder, metabolic acidosis, gastrointestinal and dermatological disorders. It is vital that patients are prepared for renal replacement therapies at CKD stage 5 which include timely education including nutritional support along with psychological and social support.

KEY WORDS

- ♦ Chronic kidney disease ♦ Cardiovascular disease
- ♦ Patient education ♦ Nursing care

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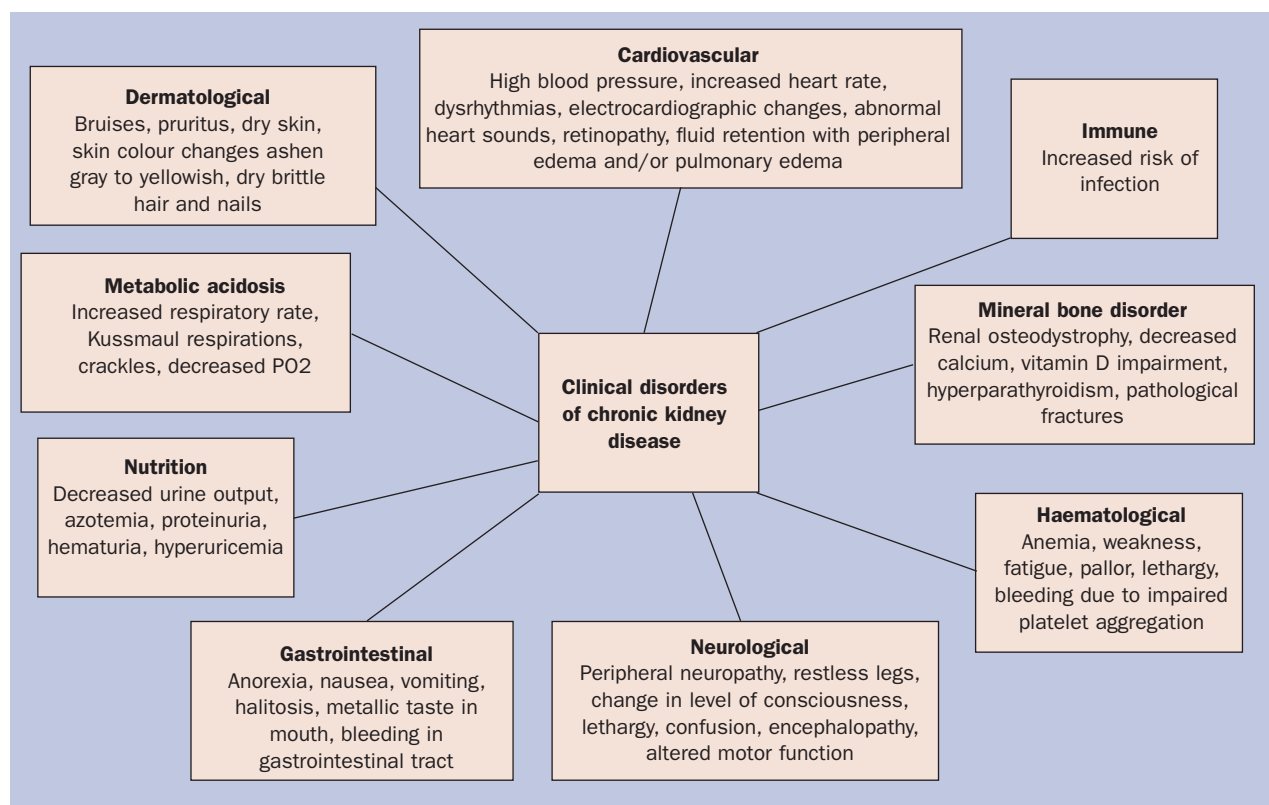


Figure 1. Clinical disorders of chronic kidney disease (adapted from Brosious and Castagnola, 2006)

1 year, or more than 10 ml/minute/1.73 m² in 5 years. It is important to pay particular attention to those individuals in whom a reduction in GFR continues at a rate which would (extrapolating the current pace of decline) lead to them needing renal replacement therapy within their lifespan (NCC-CC, 2008).

CVD remains the most frequent source of death in patients with end-stage renal disease (ESRD), and CKD patients are most likely to die from CVD than are expected to advance to ESRD. There is overwhelming scientific evidence to support the facts that proteinuria and decreased GFR are independent risk factors for CVD (Figure 2). The CKD population has an increased prevalence of traditional cardiovascular risk factors such as diabetes, hypertension, and dyslipidaemia (Chonchol and Spiegel, 2009) (see Figure 3 for traditional and non-traditional risk factors associated with CKD).

Hypertension

It is vital that hypertension is treated to slow the downward spiral of CKD. There are many benefits with regards reducing blood pressure, including the risk of a stroke being decreased by 35–40%, the risk of a myocardial infarction (MI) being decreased by 20–5% and the risk of heart failure decreasing by 50% (Benavente et al, 2008a). NICE guidelines (NCC-CC, 2008) recommend aiming to keep the systolic blood pressure below 140 mmHg (target range 120–139 mmHg) with the diastolic blood pressure below 90 mmHg for individuals with CKD. For those patients with CKD and diabetes and those that have an albumin creatinine ratio (ACR) of 70 mg/mmol or more,

the recommendation is to keep the systolic blood pressure below 130 mmHg (target range 120–129 mmHg) and the diastolic blood pressure below 80 mmHg.

The first choice of antihypertensive drugs should be ACE inhibitors followed by ARBs if the ACE inhibitors are not tolerated (NCC-CC, 2008). These medications reduce systemic blood pressure and also reduce intraglomerular pressure thereby protecting the kidneys from more severe disease progression. Both the serum creatinine and potassium should be monitored within 1 to 2 weeks after commencing a patient on an ACE inhibitor or an ARB. It may be necessary to repeat this monitoring with dose increases, especially in individuals more at risk of hyperkalaemia (Young, 2008).

The role of the nurse is paramount here through health promotion. Lifestyle modifications for patients to manage hypertension include weight reduction, dietary sodium reduction, physical activity and moderations of alcohol consumption (Benavente et al, 2008a). Patients should be educated regarding the importance of recording blood pressure monitoring and be encouraged to carry out blood pressure monitoring at home (Young, 2008).

Dyslipidaemia

Dyslipidaemia can occur early in CKD and can deteriorate as the overall kidney function declines which is the case at stages 4 to 5 CKD. There are various contributing factors including the presence of diabetes, proteinuria and the level of kidney function. As patients with CKD are at the highest risk for CVD, it is wise to treat dyslipidaemia aggressively (Benavente et al, 2008a). However, the NICE

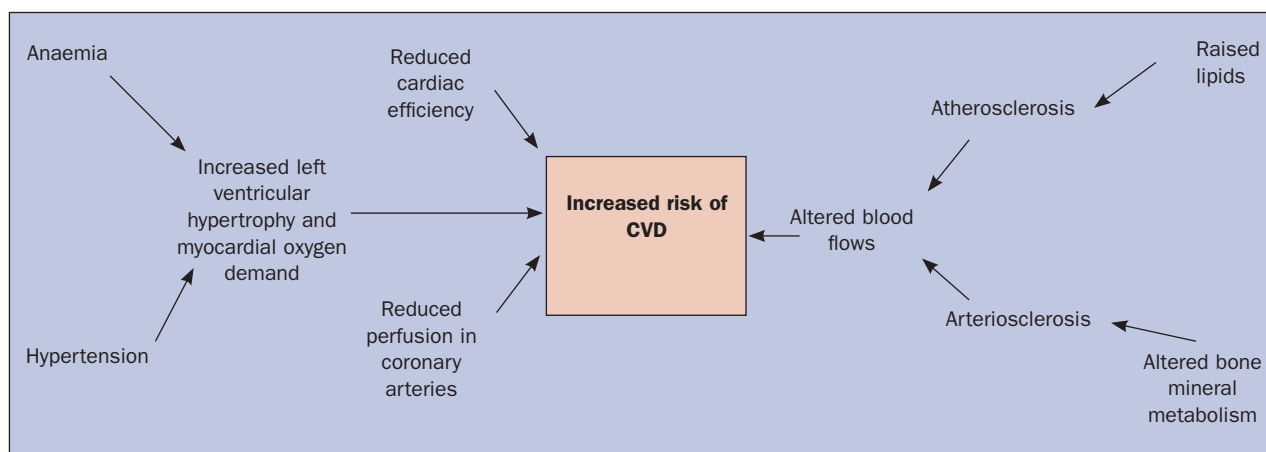


Figure 2. Some postulated pathophysiological links between CKD and increased CVD risk (Burrows and Muller, 2007)

guidelines (NCC-CC, 2008) suggest that the use of statin therapy for the primary prevention of CVD in individuals with CKD should not differ from its use in individuals without CKD and should be reflected by existing risk tables for individuals with and without diabetes. They further identify that statins should be offered to individuals with CKD for the secondary prevention of CVD irrespective of baseline lipid values. Adil and Aslam (2009) assert that patients with CKD should generally receive a statin if they have elevated levels of low density lipoprotein (LDL) if dietary manipulation is unsuccessful after 3 months.

Anaemia

Individuals with CKD stages 4 and 5 are likely to have renal anaemia and should gain benefit from correction of their anaemia (Bennett, 2008). Brown et al (2007) identify that increasing the haemoglobin (Hb) at these stages can assist in alleviating many uraemic symptoms such as cog-

nitive function, nutrition, sleep patterns and the improvement in exercise capacity. Anaemia, CKD and CVD are closely related in their pathogenesis and development. Anaemia begins early in CKD and can be linked with the decline in GFR and the development of CVD (Rao and Pereira, 2005). Silverberg et al (2003) suggest the concept of a cardiorenal anaemia syndrome which highlights that anaemia, CKD and congestive heart failure form a self-perpetuating triad as a result of the interactions amongst them resulting in multiple mortality (Figure 4).

It is well recognized that CKD predisposes a patient to CVD and can ultimately lead to congestive heart failure. Congestive heart failure and anaemia are also themselves major causes of the decline of kidney function. Anaemia can hasten the advancement of kidney disease through two mechanisms: directly by kidney tissue hypoxia and indirectly by inducing cardiac failure, reduction in the blood flow of the kidney, leading to a reduction in tissue perfusion and oxygen delivery (Rao and Pereira, 2005). The patient must be screened for anaemia (Table 3).

There are various international and country-specific standards and guidelines that exist for anaemia management in CKD (Murphy et al, 2008a). The National Kidney Foundation Kidney Dialysis Outcome Quality Initiative (NKF-K/DOQI) (2007) recommends that the Hb target should usually be in the range of 11.0–12.0 g/dl and should not be more than 13.0g/dl for both non dialysis and dialysis individuals with CKD receiving treatment with erythropoietin-stimulating agents (ESAs) (some examples being epoetin alfa and darbepoetin alfa).

NICE guidelines on anaemia management in CKD (NCC-CC, 2006) recommend an Hb target of 10.5–12.5 g/dl for adults and children over 2 years of age with CKD. It is imperative that individuals at stages CKD 4 and 5 are all iron-replete so that they can achieve and maintain this target Hb level whether or not they are receiving ESAs. Patients must be closely monitored when receiving ESAs; this includes blood pressure along with regular monitoring of iron status (Bennett, 2008).

There is abnormal haemostasis in CKD which is associated with prolonged bleeding time, including decreased activity of platelet factor III, abnormal platelet aggregation

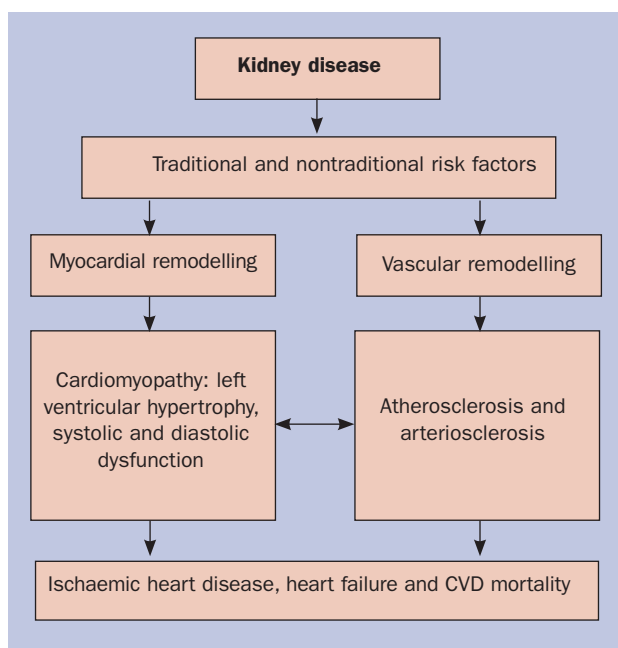


Figure 3: Traditional and nontraditional risk factors associated with CKD (Menon et al, 2005)

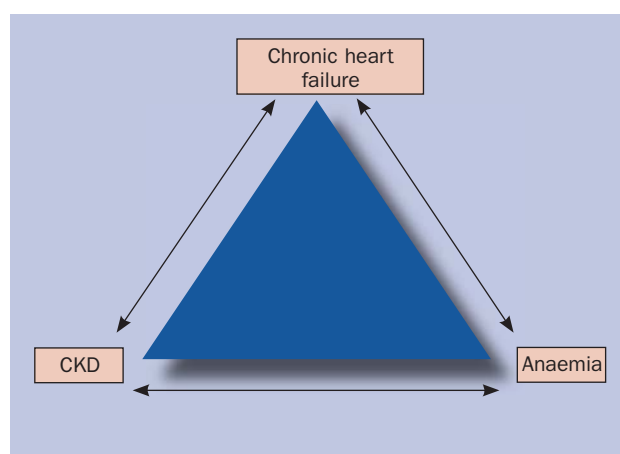


Figure 4. Cardiorenal anaemia syndrome self-perpetuating triad. Rao and Pereira (2005)

and adhesiveness and impaired prothrombin consumption. This can manifest in a tendency for abnormal bleeding and bruising, including bleeding into the pericardial sac, bleeding from surgical wounds and also gastrointestinal bleeding which can result in haematemesis or melaena. Despite these bleeding tendencies individuals with CKD have a greater tendency to thromboembolic complications (Benavente et al, 2008a).

Uraemic toxins inhibit erythropoiesis and haemolysis as a result of the uraemic changes to the red cell membrane (Hurst and Thomas, 2008). The clinical manifestations of uraemic syndrome (presence of renal failure, metabolic abnormalities such as hyperkalaemia, hyperparathyroidism, anaemia and hypertension) are usually seen after the eGFR falls below 10 ml/minute although some patients may be symptomatic at higher eGFR levels such as those with diabetes. The symptoms include nausea, vomiting, anorexia, fatigue, weight loss, pruritus and alterations in mental status. There is no one uraemic toxin that causes all of the clinical symptoms of uraemia. However, dialysis must be initiated with the presence of uraemic syndrome

**Table 3.
Screening for anaemia**

- ◆ Full blood count
- ◆ Reticulocyte count
- ◆ Serum ferritin
- ◆ Serum % transferrin saturation
- ◆ % hypochromic red cells
- ◆ Serum B₁₂
- ◆ Red cell folate concentration
- ◆ C-reactive protein (CRP)
- ◆ Tests for haemolysis
- ◆ Assessment of occult gastrointestinal blood loss
- ◆ Nutritional status

Bennett, 2008

(Guo and Aslam, 2009). Uraemic pericarditis can also be seen in 6-10% of patients with advanced CKD who are not yet receiving dialysis treatment which may be due to uraemic milieu or viral infection. Symptoms include precordial chest pain, dyspnoea and fever (Guo and Aslam 2009).

CKD affects three types of cells in the blood including erythrocytes, platelets and leukocytes. This results in anaemia and bleeding tendencies (which have already been discussed) and impaired immune and inflammatory responses. Patients must be assessed for signs and symptoms of an impaired inflammatory and infectious response as infection can be a common occurrence with this illness and can lead to morbidity and mortality (Broscious and Castagnola, 2006). Patients with CKD should be vaccinated for pneumonia and influenza. They should be screened for hepatitis B antigen and antibody and if these results are negative a vaccination along with a follow-up test for antibody is required in order to reduce patients' exposure to blood-borne pathogens in dialysis units along with the identification of carriers of the hepatitis B virus (Dinwiddie et al, 2006). There should be monitoring of hepatitis C and HIV in some dialysis units before commencing renal replacement therapy (Murphy et al, 2008b)

Potassium disorders

The ability of the kidneys to maintain potassium at near normal levels is usually maintained until the GFR falls to less than 20–25 ml/minute or individuals become oliguric. Hyperkalaemia can occur in CKD as a result of non-adherence to dietary restrictions, infection, constipation, following episodes of acute illness, haemorrhage, blood transfusion and medications such as ACE inhibitors, beta-blockers, ARBs, potassium-sparing diuretics and nonsteroidal anti-inflammatory drugs (NSAIDs).

It can be aggravated also by salt and water depletion, an extracellular shift of potassium which can occur in chronic metabolic acidosis or a lack of insulin. The management of hyperkalaemia conservatively includes attention to diet such as the avoidance of potassium-rich foods such as chocolate, bananas and citrus fruits and avoidance of the above mentioned medications.

However, for those patients taking ACE inhibitors or ARBs these should be temporarily stopped or a loop diuretic or oral sodium bicarbonate added to the regime often ameliorates hyperkalaemia. Renal replacement therapies such as haemodialysis are indicated in refractory hyperkalaemia (Benavente et al, 2008a; Hurst and Thomas, 2008; Guo and Aslam, 2009).

CKD mineral and bone disorder

The kidneys play a pivotal role in the management of calcium and phosphorus along with the activation of vitamin D by hydroxylating 25-hydroxyvitamin D to calcitriol (Burrows and Muller, 2007). In patients not yet receiving dialysis it is evident that histological abnormalities of bone begin early in the stages of CKD. There are alterations from secondary hyperparathyroidism and almost no patient on

dialysis has normal bone histology (Moe, 2006). The Kidney Disease Improving Global Outcomes (KDIGO) position statement identifies CKD mineral and bone disorder (CKD-MBD) as a broad clinical disorder that develops in any one or a combination of a number of manifestations (Moe et al, 2006) (*Table 4*).

During stages 3 and 4, calcium levels reduce significantly, while phosphate levels progressively increase in stages CKD 4 and 5. Calcium phosphate product may increase at stage 4 CKD and advance during stage 5 and with time on dialysis. Parathyroid hormone (PTH) levels increase early and significantly until stage 5 often leading to advanced secondary hyperparathyroidism (SHPT). The decreasing levels of vitamin D can occur early in CKD which can contribute to the elevation of PTH. Non-traditional CVD risk factors must be considered in CKD and these include MBD (Limrick and Delaney, 2008). Moe et al (2006) identify that MBD in CKD can result in cardiovascular calcification, morbidity and mortality along with a reduced quality of life.

NICE guidelines recommend that serum calcium, phosphate and PTH concentrations in individuals at stages 4 to 5 CKD should be monitored but the subsequent frequency of monitoring should be determined by the measured values and the clinical circumstances (NCC-CC, 2008). NKF-K/DOQI (2003) advises that PTH, calcium and phosphate should be measured every three months for patients at stage CKD 4, and at stage 5 these bloods should be measured every month.

CKD MBD is often asymptomatic, or the symptoms appear late in the process. Non-specific symptoms can include pain and stiffness in the joints, predisposition to fracture, proximal muscle weakness and spontaneous tendon rupture. Extraskeletal involvement is also present in MBD; some examples being cardiovascular calcification and calciphylaxis. Vascular calcification is accelerated in CKD with excess calcium deposited into blood vessels, the myocardium and cardiac valves. There is also a strong association with accelerated risk of stroke and amputation. Arterial calcification is linked with valvular heart disease, myocardial infarction, congestive heart failure, endocarditis, and death (Benavente et al, 2008a).

The management of CKD MBD includes attention to diet, i.e. the reduction of an abnormally high consumption of foods rich in phosphate such as dairy products and processed meats. Oral phosphorous binders can be pre-

scribed if dietary reduction is insufficient such as sevelamer hydrochloride (renagel). It is important to keep the serum calcium towards the middle or low range of normal, to reduce the risk of vascular calcification. The administration of active vitamin D sterol has been associated with improved survival and better cardiovascular outcomes (Pendse and Singh, 2007). Dinwiddie et al (2006) assert that nurses should provide information to patients with regard to CKD MBD (such as that already discussed) along with weight-bearing exercises, fall prevention, and the avoidance of NSAIDs (which are nephrotoxic). Other risk factors should be addressed such as osteoporosis, muscle strength and gait, presence or absence of bone pain and any reduction in movement.

Metabolic acidosis

The ability of the kidneys to effectively eliminate the total acid load is reduced with advancing CKD (Young, 2008). There is a decrease in total ammonia excretion which is responsible for acidosis. This results in a decreased net hydrogen secretion and a reduction in serum bicarbonate. The adverse consequences of this include protein malnutrition, suppression of albumin synthesis. There is also dissolution of bone leading to deterioration uremic osteodystrophy as a result of the net positive acid balance. Early treatment is recommended such as oral bicarbonate therapy, however, care must be taking not to hasten or worsen hypertension with the added sodium intake (Chonchol and Spiegel, 2009). Hurst and Thomas (2008) and Benavente et al (2008a) suggest that the safest method at this stage to control bicarbonate balance is dialysis.

Gastrointestinal disorders

Anorexia, weight loss, nausea and vomiting, halitosis, a metallic taste in the mouth, hiccups, are frequent findings as the patient advances towards ESRD. Ammonia, which is produced from urea broken down in the gut, acts as an irritant. There is also increased gastric acid secretion from increased release of PTH. The side effects of some medications like oral phosphate binders and oral iron medication can cause heartburn. Constipation is another potential effect, due to a reduced fluid intake as CKD advances in stages 4 and 5. Overall the individual at these stages can be less physically active and therefore more lethargic and tired (Broscious and Castagnola, 2006; Hurst and Thomas, 2008). The role of the nurse is to assess individuals' oral mucous membranes, monitor weight, observe stools for occult blood, monitor breath odour (Broscious and Castagnola, 2006) and to liaise with the renal dietician.

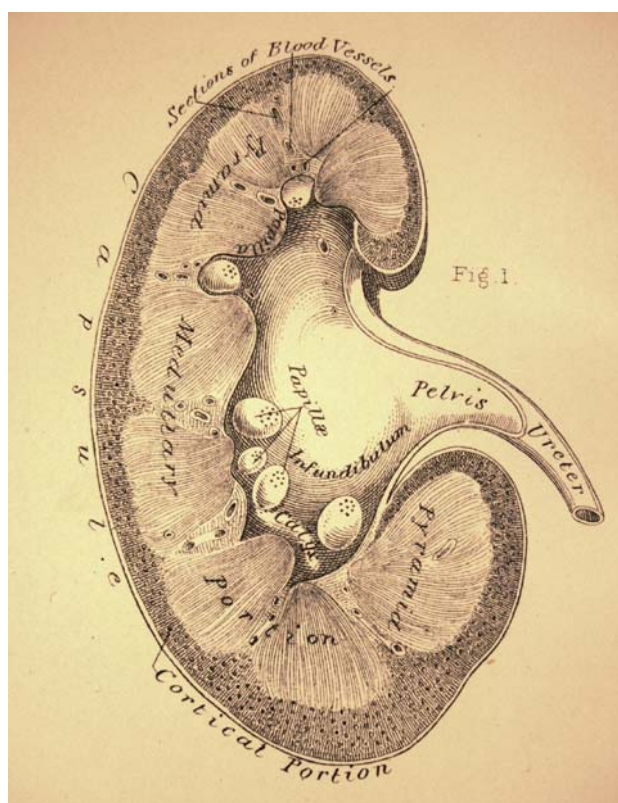
Dermatological disorders

As CKD advances there is deposition of calcium and phosphorus in the skin, and the sweat glands excrete urea resulting in uraemic pruritis. Other dermatological side effects include bleeding and bruising of the skin as a result of platelet defects and capillary permeability, particularly if the patients are constantly scratching. Individuals at stages 4 to 5 may also present with a yellow tinge to their

Table 4.
Pathophysiology of CKD

- ♦ Abnormalities of calcium, phosphorous, parathyroid hormone, and vitamin D metabolism
- ♦ Abnormalities of bone turnover, mineralization, volume, linear growth and strength
- ♦ Vascular and soft tissue calcification

Moe et al, 2006



skin due to a combination of hypermelanosis and the deposits of pigments in the skin that are generally excreted by the kidneys. There is antihistamine medication available which should assist with pruritis; however, the long term relief would be the commencement of dialysis (Hurst and Thomas 2008).

Nutritional support

The UK's Renal Association (2007) identifies the need for patients to receive dietary education at stage 4 and 5 CKD. Appropriate dietary intervention at stage 4 CKD can improve both symptomatic and biochemical control while retaining good nutritional status. Nutritional involvement is vital for those patients with ESRD or at stage 5, in order to achieve optimal fluid balance, biochemical control, continuing long term disease management and sustaining or improving overall nutritional status (Alderdice and Poulia, 2008). Patients must be individually managed using appropriate assessment tools which include reviewing clinical history, biochemical data, anthropometry and dietary assessment (Alderdice and Poulia, 2008).

There remains controversy over restriction of protein in order to slow the progression of CKD (Benavente et al, 2008b). Pendse and Singh (2007) assert that one needs to be prudent when restricting protein intake, particularly in malnourished CKD patients. Those patients who are malnourished when commencing dialysis have a poorer survival rate than their well-nourished counterparts. NKF-K/DOQI (2000) recommends that dietary protein should be 0.6 g/kg/day in stages 4 and 5 with the emphasis on high biological value of protein. The caloric intake should be 30–35 kcal/kg per day (slightly less for patients over

60 years of age). There should be close monitoring for any evidence of malnutrition, either by clinical parameters or by serum albumin as failing nutritional status is a key determinant for commencing renal replacement therapy (Pendse and Singh, 2007).

Hyperkalaemia raises the danger of sudden cardiac death. Serum potassium should be measured and potassium dietary intake reduced (such as avoiding potatoes and bananas) if abnormal laboratory values are noted. Dietary phosphate intake should be limited to 800–1000 mg/day, while calcium intake should not exceed 2000 mg. Fluid intake should only be reduced if there is evidence of oedema or fluid retention (Alderdice and Poulia, 2008). Loop diuretics such as furosemide can assist with sodium excretion from the kidneys and thus prevent sodium retention (Hurst and Thomas, 2008). Engel (2008) identify that when a patient becomes oliguric and ultimately anuric, the intake of salt and fluid will need to be restricted to control interdialytic weight gain with haemodialysis treatment and fluid balance with peritoneal dialysis.

The issue of obesity is important when assessing CKD patients. It is important that the diet is low in fat, given the strong association between CKD and CVD. In patients who are obese and are non-uremic there is an association with increased death rate along with CVD risk and inflammatory mediator levels. Marked obesity may also increase the degree of both proteinuria and advance the rate of progression of renal disease (Pendse and Singh, 2007).

Those patients who are diabetic and at these stages of CKD should be encouraged to eat starchy carbohydrate and polyunsaturated oils and fats as energy sources. It may also be essential to relax diabetic guidelines for a small period of time if the patients' appetite and intake of oral fluids reduces as ESRD advances (Alderdice and Poulia, 2008). Murphy et al (2008b) identify that these diabetic patients who require dialysis may also need insulin adjustments as the glucose content of dialysate and peritoneal dialysis solutions can influence the blood sugar levels. Teasdale (2008) asserts that the two key aims of treatment in diabetic patients at stages 4 and 5 CKD include the prevention of additional damage from micro and macrovascular disease. It is also important to ensure adequate dialysis treatment or recovery from transplant surgery depending on the renal replacement therapy choice.

Timely education

Several studies have highlighted that late referral to nephrologists is common and is associated with poor outcomes such as increased morbidity and mortality, more severe uraemia, increased use of percutaneous vascular access with associated morbidity, reduced use of the preferred arteriovenous fistula for vascular access, limited patient choice of treatment modality, longer and more costly hospitalisations at the start of dialysis, and higher rates of emotional and socioeconomic problems (Chonchol and Spiegel, 2009). The Renal Association (2007) identifies that in order to facilitate the preparation of the management of established renal failure and treatment of CKD

associated complications; those patients who are at stage 5 and those that are at stage 4 who are expected to progress to stage 5 should be followed up at a clinic. This clinic should offer counselling concerning treatment modalities and transplantation along with dietary education and broad management of anaemia from at least six months prior to the start of established renal failure. It is important to note that not all patients with CKD stage 4 will progress to stage 5; however, their risk of progression should be assessed (Renal Association, 2007).

The NICE guidelines (NCC-CC, 2008) assert that health professionals should have both the specialist knowledge and the necessary skills to facilitate learning when providing information and education programmes about CKD. They should take account of the psychological aspects of coping with this chronic illness and provide access to the appropriate support such as peer support groups, counselling or specialist nurses. The manner in which patient education is imparted by health professionals can impact upon patients' decisions regarding treatment modalities. All treatment options: haemodialysis, peritoneal dialysis, transplantation and not having dialysis should be regarded as equal along with the importance that each patient must be assessed individually (Murphy et al, 2008b).

Issues such as drug regimen, nutrition, modality choice, vascular access placement, lifestyle, self-management, employment and rehabilitation should also be discussed at stage 4 CKD. Information on palliative or hospice care should also be presented and discussed with the patient especially if he or she has significant co-morbidities. If a patient chooses not to undergo or to withdraw from dialysis treatment, every body system will become increasingly dysfunctional resulting in worsening uraemia including pulmonary oedema, restlessness and confusion. The aim of palliative care is to manage these symptoms and allow for a peaceful death for the patient and his/her family members (Dinwiddie et al, 2006).

Patients that are affected by CKD are required to make vast changes to their lives that include a life-long commitment to a treatment that does not provide cure and is palliative in nature (it prolongs life and reduces kidney disease symptoms). This does not take account of the sheer extent of the decision making processes that patients must face such as what treatment option to choose in the first instance. Continual important decisions need to be made as life and circumstances change such as whether to continue to receive dialysis treatment, or consent to transplantation (Bevin, 2008), or to withdraw from dialysis.

There are various emotional and physical consequences of living with advanced CKD. There can be relationship difficulties including sexual difficulties caused by issues such as uraemia and loss of libido. There are many difficulties with complying or adhering to renal replacement therapy which will be discussed in the next article. Patients faces many struggles with this life changing event, not alone do they have to face the challenges of living with renal replacement therapy but so do their families/carers/significant others.

KEY POINTS

- ◆ CKD is classified into five stages with many of the risk factors for CVD are also risk factors for CKD
- ◆ The nursing care of patients with CKD stages 4 to 5 include the management of the many physical symptoms and the preparation for renal replacement therapies which includes timely education along with psychological and social support
- ◆ Nurses play a vital role in the education of these patients at the advanced stages of this chronic illness

This article addressed CKD stages 4 and 5 and its inter-relationship with CVD. There are many physical symptoms that impact on patients at advanced CKD. It is important to manage these symptoms. Nurses must maintain their vital role in promoting healthy lifestyles for patients at these stages and encourage them to become involved in their care. The final article in this series will address renal replacement therapies and their interrelationship with CVD.

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